

Thickness reconstruction of nuclear power plant pipes with flow-accelerated corrosion damage using laser ultrasonic wavenumber imaging

Structural Health Monitoring
2018, Vol. 17(2) 255–265
© The Author(s) 2017
Reprints and permissions:
sagepub.co.uk/journalsPermissions.nav
DOI: 10.1177/1475921716689733
journals.sagepub.com/home/shm


Thanh Chung Truong and Jung-Ryul Lee

Abstract

The pipeline is an essential part of nuclear and other power plants. Designed to operate for decades, the pipeline is susceptible to multiple types of damage due to working in a high pressure, high temperature, and highly corrosive environment. Therefore, it requires regular inspection. The issue of many current nondestructive testing systems for pipelines is the use of a one-dimensional scan mechanism along the longitudinal direction of the pipe, which is incompatible with the two-dimensional curvature of elbow pipes, yet it is at these places in which the flow changes directions inside the elbow pipes that a common type of damage occurs: flow-accelerated corrosion damage. In this article, flow-accelerated corrosion damage is visualized using the Ultrasonic Wavenumber Imaging algorithm, which maps the dominant local spectroscopic wavenumber of the wavefield at a particular mode and a particular frequency. In previous studies, the mode and frequency were chosen empirically. In this article, a novel theoretical study is presented to optimize the two input parameters of Ultrasonic Wavenumber Imaging based on a sensitivity analysis of the spectroscopic wavenumber with respect to the change in thickness of the structural base. Additionally, an inverse method is proposed to estimate the thickness of the structure based on the Ultrasonic Wavenumber Imaging map.

Keywords

Power plant pipes, elbow pipe inspection, flow-accelerated corrosion, ultrasonic propagation imager system, Ultrasonic Wavenumber Imaging algorithm, thickness reconstruction

Introduction

Today, the dominant technology for inspection of pipelines in nuclear and other power plants is the ultrasonic C-scan system, where a variety of products can be found on the market supporting manual control or robotic control via a handheld device. The principle of the C-scan system is well known: the ultrasonic waves are generated and sensed through the inspected structure to provide an image of the damages that reflect and scatter ultrasonic waves within and on the surfaces of the inspected structure. However, the conventional C-scan has several limitations that prevent it from being adopted more widely. First, in practical applications, the C-scan is still a laborious process. Second, it requires direct contact between the transducer head and the pipe structure, which is undesirable in some cases due to high temperature or irradiation. Third, a couplant is often needed during the scanning to

overcome the mismatch of acoustic impedance between air and solid, and finally, many current nondestructive testing (NDT) systems have a one-dimensional scan mechanism along the longitudinal direction of the pipe, which is incompatible with the two-dimensional curvature in elbow pipes, yet it is at these places in which the flow changes direction inside the elbow pipes that a common type of damage occurs: flow-accelerated corrosion (FAC) damage. It is because in the case of the elbow pipe, the flow pattern is subject to large changes in flow

Department of Aerospace Engineering, Korea Advanced Institute of Science and Technology, Daejeon, Republic of Korea

Corresponding author:

Jung-Ryul Lee, Department of Aerospace Engineering, Korea Advanced Institute of Science and Technology, 291 Daehak-ro, Yuseong-gu, Daejeon 34141, Republic of Korea.
Email: leejr@kaist.ac.kr

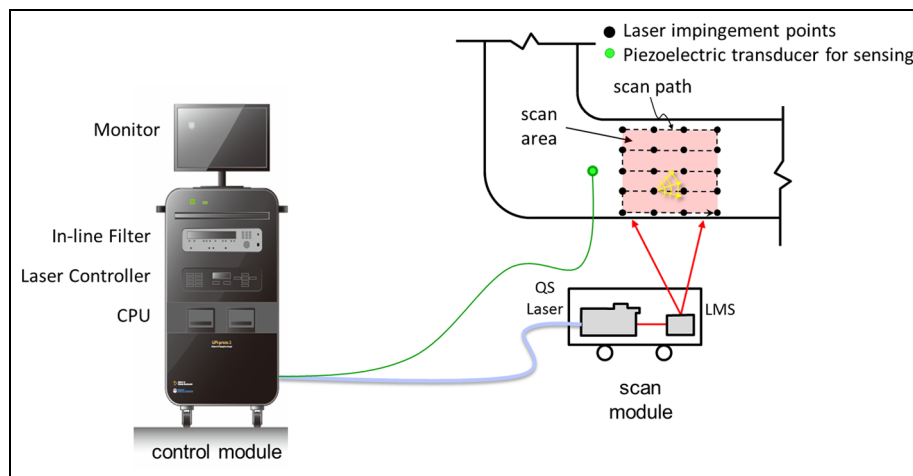


Figure 1. Ultrasonic propagation imager (UPI) for inspection of the nuclear power plant (NPP) pipeline (LMS: laser mirror scanner; CPU: central processing unit; see <http://oesl.kaist.ac.kr/xndt> for system details).

direction and flow velocity,¹ thus significantly increase the corrosion behavior.^{2,3} Several accidents were reported due to FAC that had caused leakage or rupture.⁴⁻⁶

Recently, an alternative solution for pipe inspection is being extensively investigated: full-field scanning ultrasonic systems. The term “full-field” refers to the set of data (displacement, velocity, etc.) taken on a dense grid of points in the area of interest measured during a time window. This technique has the flexibility to scan the structure using a two-dimensional (2D) scanning mechanism, such as using a mechanical stage⁷ or mirror scanner.⁸ For a more complete review on full-field scanning techniques, readers can refer to the study by Ostachowicz et al.⁹ The fundamental difference between the full-field scanning ultrasonic system and the conventional C-scan system is the measuring signal; in the former system, it is propagated by guided and surface waves, such as Lamb waves or Rayleigh waves, instead of bulk waves, as in the latter. The use of Lamb waves to detect corrosion and thinning for pipe structures has been previously reported in works by Pei et al.,¹⁰ Cawley and Alleyne,¹¹ and Bingham and Calvo.¹² In this article, a full-field scanning laser ultrasonic system, the so-called ultrasonic propagation imager (UPI),¹³ is used for inspection of pipes in a nuclear power plant (NPP), as shown in Figure 1. The UPI consists of a control module and a scan module. The control module consists of a CPU, digitizer, inline filter/amplifier, and laser controller. The scan module consists of a 2D laser mirror scanner (LMS) for scanning a Q-switched laser beam for the generation of laser ultrasonic which is typically in the thermal elastic regime. An amplifier-integrated piezoelectric transducer is used for measuring the laser ultrasonic waves. Through reciprocity, a virtual laser ultrasonic wavefield is generated

from the sensing point. This approach is different with the scanning systems using laser Doppler vibrometers¹⁴ (LDV), where a laser beam is used for ultrasonic waves sensing at various locations in the scan area, and a piezoelectric transducer or a pulse laser is used for excitation at one fixed location. In theory, the two approaches are equivalent due to the reciprocity principle in linear elasto-dynamic medium. However, in practice, the approach chosen for the UPI has the following advantages: the laser excitation allows a wide range of beam incident angle,¹⁵ whereas the signal of the LDV is significantly altered when the incident angle changes; also in the case of using LDV for sensing, surface preparation is often needed to ensure good-quality signals.

Given the advantages of speed and convenience, one trade-off of using the full-field scanning ultrasonic system for structural health management is the relative complexity of signal processing due to the multi-mode and dispersive property of surface waves. For that reason, many algorithms have been developed for damage visualization using the full-wavefield data. One way to classify these algorithms is by looking at what domain the wavefield data are mainly processed for damage visualization: in time domain, frequency domain, or wavenumber domain. Some examples of time-domain algorithms are as follows. Lee et al.¹⁶ proposed the adjacent waves subtraction algorithm to suppress incident waves and exaggerate anomalous waves, such as by confining and scattering waves caused by damages. The main assumption of this algorithm is that the waves from two adjacent scanning points are very similar except in the areas where structural anomalies, such as damages, occur. Sohn et al.¹⁷ extracted the confining waves in the delamination site of a multilayer composite specimen and visualized the delamination using the

cumulative kinetic energy index. A similar index was proposed by Ruzzene et al.,¹⁸ the so-called root-mean-square index. Some examples of frequency-domain algorithms are as follows. Chia et al.¹⁹ proposed mapping the frequency component of the ultrasonic waves to visualize damage. The principle behind this algorithm is that when incident ultrasonic waves encounter structural anomalies, the frequency component of the scan points around the damages changes. The change is due to the reflected, diffracted, mode-converted, and scattered waves. Rogge and Johnston²⁰ used the wavelet transform coefficients as damage indicators. Some examples of wavenumber-domain algorithms are as follows. Ruzzene²¹ applied a directional wavenumber filter to the wavefield to remove the incident waves and highlight the reflection waves associated with damages. Michaels et al.²² developed an algorithm for mode separation and incident wave removal to detect delamination in a composite plate and a crack emanating from a hole on an aluminum plate. Yu and Tian²³ used a short-space 2D Fourier transform to obtain the space-frequency-wavenumber representation and studied the wave interaction with structural discontinuity on an aluminum plate. Kudela et al.²⁴ applied a mask in the wavenumber domain to remove the main component of propagating waves in the space/time domain and then calculated the cumulative kinetic energy index of the filtered wavefield. Rogge et al.²⁵ proposed using a local Fourier domain analysis method to estimate the spatially dependent wavenumber, which can be used to determine the delamination depth in composite structures. Mesnil et al.²⁶ developed a fast quantification of the wavenumber so-called instantaneous wavenumber to visualize the delamination and the corresponding effective thickness map in composite structures. Flynn et al.²⁷ isolated a single-wave mode and mapped the wavenumber at one particular frequency that maximizes the wave energy at each grid-scanning point, called local wavenumber estimation. The same method was called Ultrasonic Wavenumber Imaging (UWI) in the UPI system. Several applications of the wavenumber spectroscopy and algorithm modification can be found in the inspection of layer-by-layer 3D-printed parts^{28,29} and composite and pipe structures.^{28,30}

In this article, we further improve the UWI algorithm for a more systematic selection of image-processing parameters and we show the application of using the UWI algorithm for damage visualization of an NPP pipeline, both straight and elbow pipes. In detail, a novel theoretical study is presented to optimize two essential input parameters of the UWI algorithm (Lamb wave mode and frequency), which previously was only chosen empirically.^{27,28} This proposed technique is based on a sensitivity analysis of the spectroscopic wavenumber with respect to a change in

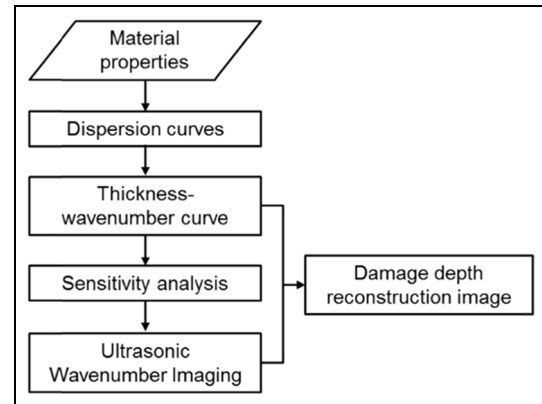


Figure 2. Damage characterization using Ultrasonic Wavenumber Imaging (UWI) algorithm.

thickness. Finally, an inverse method is proposed to estimate the thickness of the structure based on the UWI map.

This article is organized as follows. The following section details the theoretical studies, including a brief theory of the UWI algorithm and the proposed technique for selection of optimum input parameters. The next section describes the experimental cases in which the UWI algorithm is applied for damage visualization and reconstruction of the thickness of straight and elbow NPP pipes. The final section concludes with a discussion and conclusions.

Damage characterization using UWI algorithm

This section details the proposed algorithm for damage visualization and damage depth reconstruction imaging. The first part introduces the Rayleigh-Lamb equation of the dispersion curve.³¹ Using numerical methods, the Rayleigh-Lamb equation can be solved, which gives a library of dispersion curves at different thicknesses. The second part shows a sensitivity analysis of the spectroscopic wavenumber when the thickness of the structure changes. Essential parameters of the UWI algorithm (Lamb wave mode and frequency) are chosen, so the sensitivity of the wavenumber with respect to a change in thickness is at a maximum. The third part briefly presents the UWI algorithm. Readers can refer to the study by Flynn et al.²⁷ for a more rigorous mathematical treatment. The final part of this section documents the inverse method for thickness reconstruction based on the UWI map. A general flow-chart of the proposed algorithm is shown in Figure 2.

Dispersion curve

The dispersion curve is a well-known concept of wave propagation in a solid. It relates the circular

wavenumber and frequency of propagating waves. The dispersion curve of a Lamb wave is described by the Rayleigh-Lamb equation³¹

$$\frac{\tan(qh)}{\tan(ph)} = - \left[\frac{4k^2pq}{(q^2 - k^2)^2} \right]^s \quad (1)$$

where $p^2 = \omega^2/c_L^2 - k^2$; $q^2 = \omega^2/c_T^2 - k^2$; h is the half thickness of the plate; ω is the circular frequency; $k = 2\pi/\lambda$, where λ is the wavelength; k is the circular wavenumber; c_L and c_T are the longitudinal and transverse velocities, respectively; $s = \pm 1$ corresponds to the symmetric mode at +1 and anti-symmetric mode at -1.

Equation (1) can be solved numerically, such as by following the steps presented in Rose.³¹ Figure 3 shows the dispersion curves of an STS370 stainless steel plate ($c_L = 5960$ m/s, $c_T = 3230$ m/s) at different thicknesses ($d = 2$ and $h = 1, 3, 5, 10$ mm). The continuous and dotted lines represent A_0 and S_0 modes, respectively. The number in each line is the thickness, d , of the plate corresponding to that dispersion curve. In Figure 3, the spectroscopic wavenumber, $\nu = 1/\lambda$, was plotted instead of the circular wavenumber, k . For the rest of this article when referring to “wavenumber,” it is understood as the spectroscopic wavenumber.

Sensitivity analysis of wavenumber with respect to a change of thickness

The sensitivity of wavenumber with respect to a change of the thickness at frequency f_0 is defined as follows

$$S_{f_0} = \left| \frac{\Delta \nu}{\Delta d} \right|_{f=f_0} \quad (2)$$

where $\Delta \nu$ is the change of the wavenumber, and Δd is the change in thickness. To plot the sensitivity curve, first we calculate the thickness–wavenumber relationship by slicing along the frequency axis of Figure 3. The result is the thickness–wavenumber curves, as shown in Figure 4(a). It is noted that in Figure 3, only a few different thicknesses are shown for the purpose of illustration. To plot the thickness–wavenumber curves shown in Figure 4(a), a dense thickness variation ($\Delta d = 0.1$ mm), from 1 to 12 mm and at different frequencies ($f_0 = 200, 300$, and 400 kHz), was used. Next, the sensitivity curves were obtained by taking the derivative of the thickness–wavenumber curves, as shown in Figure 4(b).

Some observations can be drawn from Figure 4(b). First, when the specimen is thin, A_0 mode has superior sensitivity compared to S_0 mode. However, when the specimen is thick, which is the typical case for NPP pipes, the S_0 mode becomes more sensitive. Second, the thickness corresponding to the peak of the sensitivity in

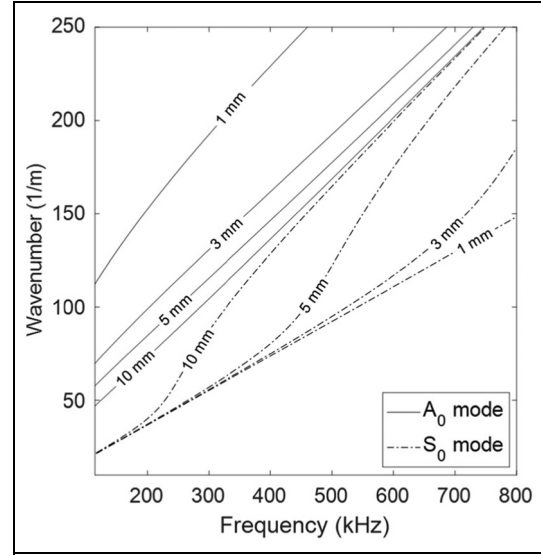


Figure 3. Dispersion curves of an STS370 stainless plate at different thicknesses ($d = 1, 3, 5$, and 10 mm). The continuous and dotted lines are the A_0 and S_0 modes, respectively. The number in each line is the thickness, d , of the plate corresponding to that dispersion curve. The horizontal axis is frequency (kHz), and the vertical axis is the spectroscopic wavenumber (1/m).

S_0 mode shifts to the right when the frequency decreases. At the same time, the amplitude of the peak also decreases. This property shows that the UWI algorithm loses effectiveness for damage visualization as the plate gets thicker. Third, for the UWI algorithm, the optimum mode and frequency should be chosen to maximize the sensitivity when the thickness is equal to the intact thickness of the specimen. For the case of the NPP pipelines investigated in this article (straight pipe with thickness 9 mm and elbow pipe with thickness 11 mm), it is apparent from Figure 4(b) that the S_0 mode is more sensitive. Subsequently, the optimum frequency is chosen by plotting the sensitivity curve of the S_0 mode at different frequencies and choosing the frequency that fixes the peak of the sensitivity curve to the intact thickness. We will clarify this point again in the next section.

UWI algorithm

In this section, a brief introduction of the UWI algorithm is presented. It is well known that the Lamb waves propagate across a plate in various modes, including symmetric and anti-symmetric modes.³¹ In the case of an intact plate with uniform stiffness, at one particular frequency of one particular mode, the guided waves have one corresponding dominant wavenumber. In the case of a plate with a damage area inside, the

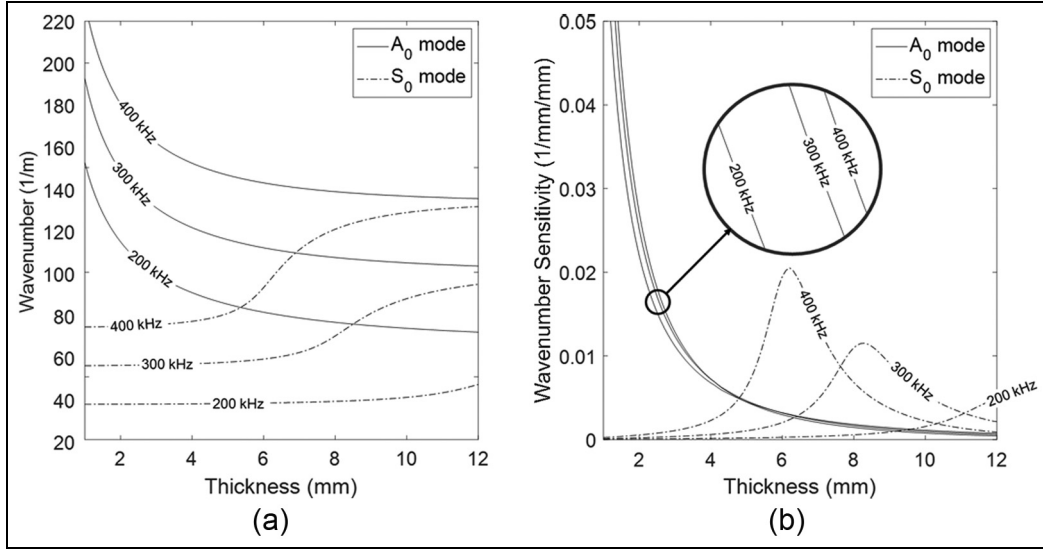


Figure 4. (a) Thickness–wavenumber curves of an STS370 stainless plate at different frequencies ($f_0 = 200, 300$, and 400 kHz) and (b) sensitivity curve of the wavenumber with respect to changes in thickness at different frequencies ($f_0 = 200, 300$, and 400 kHz), which was obtained by taking the derivative of thickness–wavenumber curves in Figure 4(a).

dominant wavenumber inside that damage area is different from the surrounding intact area. By mapping the dominant wavenumber of the wavefield at wave mode m and frequency f_0 , the damage can be visualized. That is the principle of the UWI algorithm. The flowchart of UWI is shown in Figure 5. The mathematical expressions of UWI algorithm are as follows.

To start, the three-dimensional (3D) full-wavefield matrix $u(x, y, t)$ is transformed from the space–time domain to the wavenumber–frequency domain using the 3D Fourier transform. Then, a narrow frequency bandpass filter centered at frequency f_0 is applied

$$U(\nu_x, \nu_y, f) = F_{3D}[u(x, y, t)] \quad (3)$$

$$U_{f_0}(\nu_x, \nu_y, f) = U(\nu_x, \nu_y, f) W_F(f_0) \quad (4)$$

where $W_F(f_0)$ is a Gaussian-shaped frequency domain window centered at frequency f_0 and width σ

$$W_F(f_0) = \exp\left(-\frac{(f - f_0)^2}{2\sigma^2}\right) \quad (5)$$

Next, a bank of wavenumber windows $W_N(\nu_i)$ is built and applied

$$U_{f_0, \nu_i}(\nu_x, \nu_y, f) = U_{f_0}(\nu_x, \nu_y, f) W_N(\nu_i) \quad (6)$$

where $W_N(\nu_i)$ is a Gaussian-shaped wavenumber domain window centered at the wavenumber ν_i . The result of equation (6) is a 3D wavefield in the wavenumber–frequency domain with a narrowband frequency centered at f_0 and narrowband wavenumber

centered at ν_i . An inverse Fourier transform is then applied, and the wave energy matrix $E_i(x, y)$ is calculated as follows

$$\tilde{u}_i(x, y, t) = F_{3D}^{-1}[U_{f_0, \nu_i}(\nu_x, \nu_y, f)] \quad (7)$$

$$E_i(x, y) = \sum_t [\tilde{u}_i(x, y, t)]^2 \quad (8)$$

where $E_i(x, y)$ is a 2D matrix with size $V \times H$, where V and H are the number scanning points in vertical and horizontal directions, respectively. The dominant wavenumber $\hat{\nu}(x, y)$ at the spatial point (x, y) is the wavenumber that maximizes the wave energy at that particular point of the wave energy matrices. The UWI map is obtained by mapping $\hat{\nu}(x, y)$

$$\hat{\nu}(x, y) = \arg \max_{\nu} [E_1(x, y), E_2(x, y), \dots, E_i(x, y) \dots] \quad (9)$$

Damage depth reconstruction using inverse algorithm

The wavenumber–thickness curve, as shown in Figure 4(a), shows a one-to-one relationship between the wavenumber and thickness at a given mode m and a given frequency f_0 . Using this curve, it is easy to estimate the thickness of the specimen at each point in the scan area using the UWI map. This inverse algorithm can be written as follows:

$$d_{\text{reconstruction}} = g_{\text{fit}}^{m, f_0}(\nu_{\text{UWI}}) \quad (10)$$

where ν_{UWI} is the wavenumber from UWI map, $d_{\text{reconstruction}}$ is the reconstruction thickness, and g_{fit}^{m, f_0} is

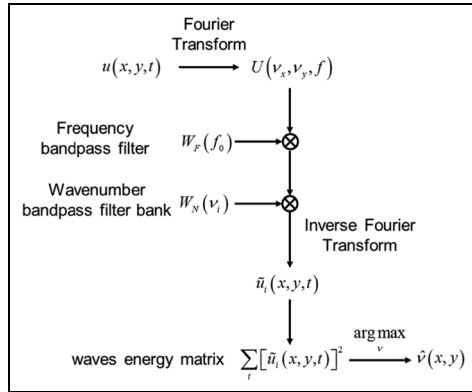


Figure 5. Flowchart of the Ultrasonic Wavenumber Imaging (UWI) algorithm.²⁷

the fitting function of the $d = d(v)$ curve at mode m and frequency f_0 .

Experiments, analysis, and results

Pipe specimens and experimental parameters

Two pipe specimens with artificial FAC damages were inspected using the UPI system. Figure 6(a) shows the picture of the straight pipe specimen. This pipe had an outer diameter of 165 mm and a wall thickness of 9 mm. FAC damage was made inside the pipe in the shape of an ellipse, as seen in the top view of the pipe in Figure 6(b). Figure 6(c) and (d) shows the sketch of two cross sections to clarify the dimensions of the straight pipe. Note that the dimensions in Figure 6(c) and (d) are not scaled. At the center of the FAC, the thickness was reduced by 60%, which resulted in the thickness of 3.6 mm. Figure 7(a) shows the picture of the elbow pipe specimen. This pipe had an outer diameter of 172 mm and wall thickness of 11 mm. Figure 7(b) and (c) shows the sketch of two cross sections to clarify the dimensions of the elbow pipe. Again, the dimensions in Figure 7(b) and (c) are not scaled. At the center of the FAC, the thickness was reduced by 27%, which resulted in the thickness of 8.03 mm.

The parameters of the UPI system used for inspection of the pipes were as follows. A Q-switched laser with a wavelength of 1064 nm was used. The energy of each pulse was set to 5.84 mJ. This is a class IV laser.¹⁵ The scan speed was set at 50 Hz, far below the maximum speed of UPI system.¹³ The severe reverberation of the metallic structure when operating at high speeds renders the signals useless. The high scan speed is suitable for structures with robust damping, such as composite structures and painted structures. The scan area sizes are shown in Figures 6(a) and 7(a); the scan area in the case of the straight pipe is 180 mm $\times$ 130 mm and

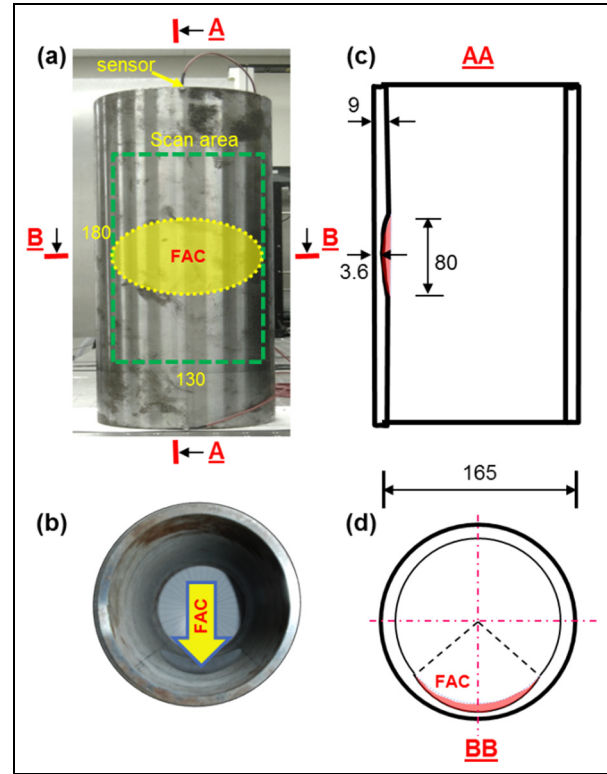


Figure 6. Straight pipe specimen: (a) front view with the scan area size, (b) top view, (c) AA cross section, and (d) BB cross section. Unit of dimensions: mm. Note that the dimensions are not scaled (FAC: flow-accelerated corrosion damage).

is 100 mm $\times$ 200 mm in the case of the elbow pipe. The distance between two adjacent points was 0.5 mm for both cases. With the settings as presented, the number of scanning points of the straight pipe and elbow pipe was 46,800 and 40,000 points, respectively, and the scanning durations of the straight pipe and elbow pipe were 936 and 800 s, respectively. The double curvature structure like the elbow pipe could be scanned because the UPI system is capable of scanning the wide range of beam incident angle of $\pm 70^\circ$. The ultrasonic waves were measured using the integrated piezoelectric sensors attached to the specimens, as shown in Figures 6(a) and 7(a).

The full scanning data form a 3D matrix $V \times H \times L$, where V and H represent the number of scanning points in the vertical and horizontal directions of the scanning area, respectively, whereas L is the sampling length of the measurement data for each scanning point. For example, in the case of the straight pipe, there are 360 points in the vertical direction, and 260 points in the horizontal direction. For both cases, the sample length is 500 points with a sampling frequency of 2 MHz, which captures 250 μ s of wave propagation. The wave propagation can be visualized by plotting the successive

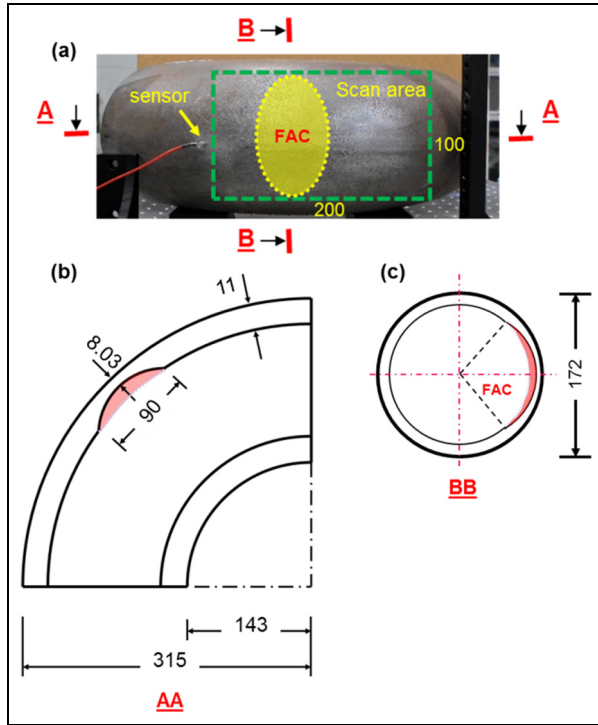


Figure 7. Elbow pipe specimen: (a) front view with the scan area size, (b) AA cross section, and (c) BB cross section. Unit of dimensions: mm. Note that the dimensions are not scaled (FAC: flow-accelerated corrosion).

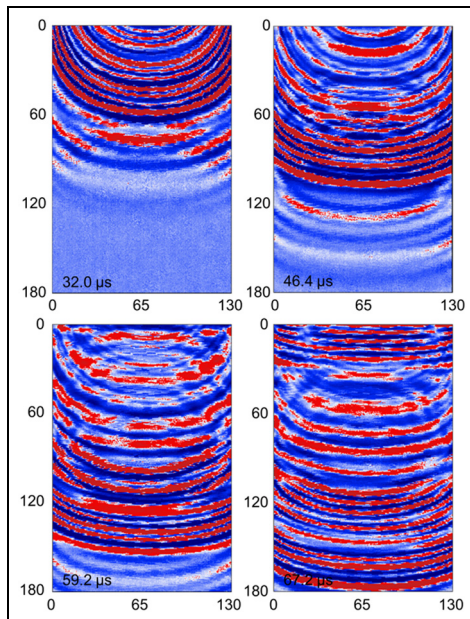


Figure 8. Freeze-frames of the laser ultrasonic wavefield of the straight pipe at $t = 32.0, 46.4, 59.2$, and $67.2 \mu s$.

slides along the time axis. Figure 8 shows the freeze-frames of the laser ultrasonic wavefield of the straight pipe at $t = 32.0, 46.4, 59.2$, and $67.2 \mu s$. Figure 9 shows

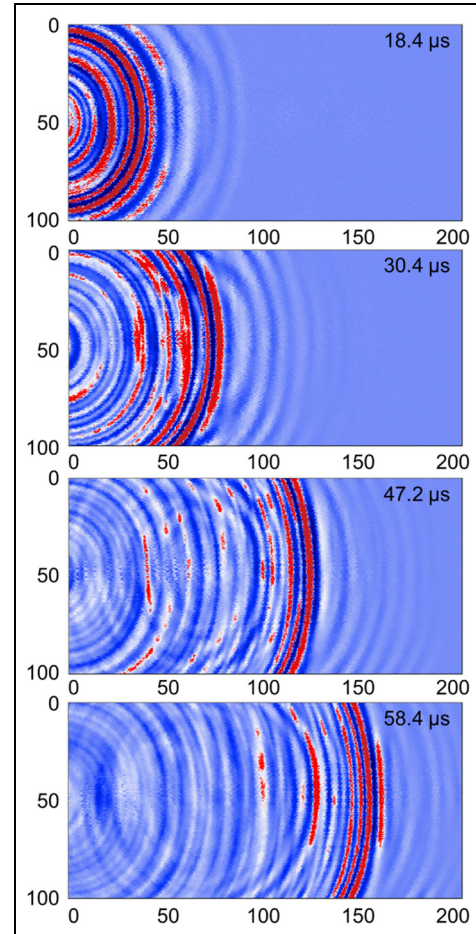


Figure 9. Freeze-frames of the laser ultrasonic wavefield of the elbow pipe at $t = 18.4, 30.4, 47.2$, and $58.4 \mu s$.

the freeze-frames of the laser ultrasonic wavefield of the elbow pipe at $t = 18.4, 30.4, 47.2$, and $58.4 \mu s$. The reflected waves from the damages can be observed in Figures 8 and 9. However, we can see that if the wave propagation video is merely used, it is difficult to characterize the FAC damages, such as estimating the size, or severity of the damages. In the next section, we will show that the UWI algorithm can be used more effectively for damage characterization.

Optimization parameters for UWI results of the straight and elbow pipes

The strategy, as presented in the previous section, is used for determining the optimum wave mode and frequency for the UWI algorithm. A frequency sweep was performed to construct a series of sensitivity curves of the wavenumber with respect to thickness. The optimum frequency is selected, so that the peak of the S_0 mode is at the thickness of an intact pipe, which is 9 mm in the case of the straight pipe and 11 mm in the

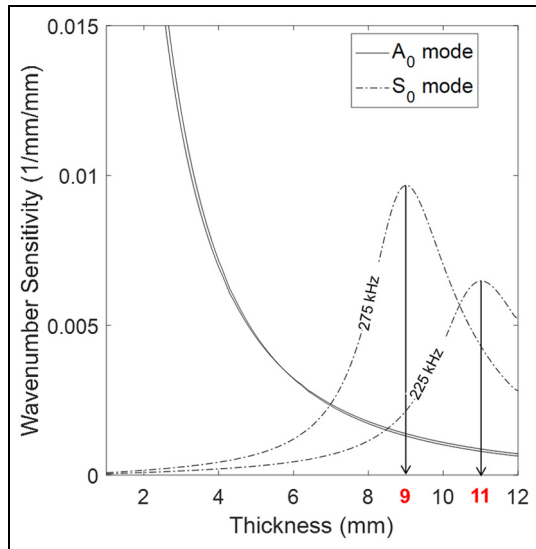


Figure 10. Sensitivity curves of the wavenumber with respect to the change of thickness at different frequencies ($f_0 = 275$, 225 kHz). The peak of the S_0 mode is at 9 and 11 mm, corresponding to the intact thickness of the straight pipe and elbow pipe, respectively.

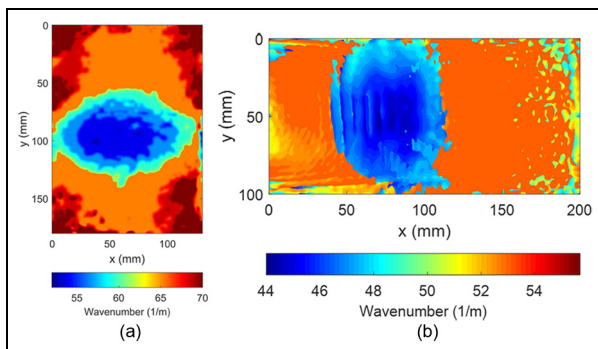


Figure 11. Ultrasonic Wavenumber Imaging (UWI) algorithm results of (a) straight pipe with input parameters: S_0 mode at 275 kHz and (b) elbow pipe with input parameters: S_0 mode at 225 kHz.

case of the elbow pipe. Figure 10 shows the optimum mode for UWI of both specimens is the S_0 mode, and the optimum frequency for generating UWI is 275 and 225 kHz for the straight pipe and elbow pipe, respectively.

UWI results of the straight and elbow pipes

Figure 11(a) and (b) shows the results of using the UWI algorithm for damage visualization of the straight and elbow pipes, respectively, with the optimized parameters obtained from the previous section. For the straight pipe, UWI was performed with the S_0 mode at

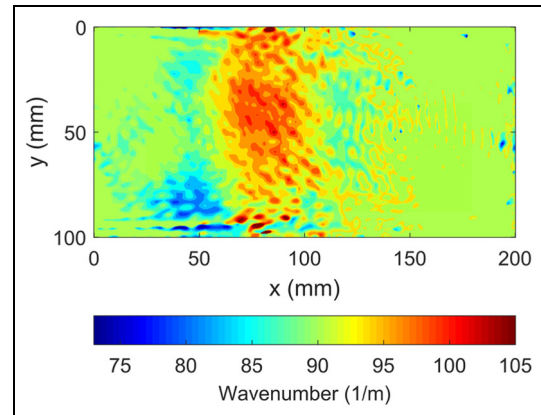


Figure 12. Ultrasonic Wavenumber Imaging (UWI) algorithm results of the elbow pipe with input parameters: A_0 mode at 225 kHz.

275 kHz, and for the elbow pipe, UWI was performed also with the S_0 mode at 225 kHz. At the intact area of the straight pipe, the wavenumber is 65.4 (1/m), which is corresponding to the wavelength of 15.29 mm. Similarly, at the intact area of the elbow pipe, the wavenumber is 55.7 (1/m), which is corresponding to the wavelength of 17.95 mm. It is clear that the spatial scan grid step of 0.5 mm is sufficient to avoid spatial aliasing issue.

A significant amount of artifact can be observed in the UWI results near the edge of the inspected area. The main reason is due to the curvature of the pipes, and the scan mechanism using mirror scanner leads to the non-uniform spatial sampling near the edges. However, the elliptical shape of the damages was captured quite clearly in both cases with UWI results. To show that the choice of wave mode is an important parameter of UWI, Figure 12 shows the UWI algorithm result of the elbow pipe with input parameters: A_0 mode at 225 kHz. It is apparent that due to the much poorer sensitivity of wavenumber due to thickness changes, the image of the damage is much less clear in this case. In the next section, we will apply the inverse method to reconstruct the damage depth using the UWI maps in Figure 11(a) and (b).

Thickness reconstruction of the pipe specimens

Figure 13 shows the thickness–wavenumber curves in S_0 mode of an STS370 stainless plate at 275 and 225 kHz. The thickness–wavenumber curve at 275 kHz is used for thickness estimation of the straight pipe, and the thickness–wavenumber curve at 225 kHz is used for thickness estimation of the elbow pipe. A minor difference between Figure 13 and Figure 4(a) is that the horizontal axis and vertical axis are switched

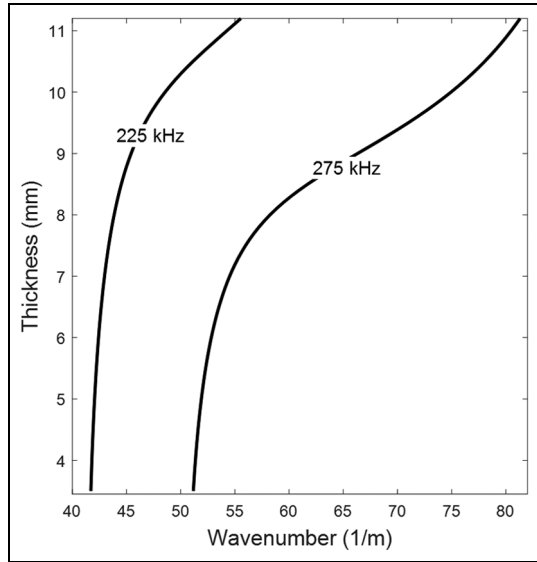


Figure 13. Wavenumber–thickness (d – v) curves of an STS370 stainless plate at 275 and 225 kHz.

to emphasize the inverse calculation to match the calculated wavenumber with the corresponding thickness.

The thickness reconstruction maps of the straight pipe and the elbow pipe are shown in Figures 14(a) and 15(a), respectively. The cross section of the thickness maps through the center of the damages are plotted in Figures 14(b) and 15(b) (line $x = 65$ mm in the case of the straight pipe and $y = 50$ mm in the case of the elbow pipe). The comparison of true and estimation of thickness is shown in Table 1. The estimation thickness was identified fairly accurately with the differences of true and estimation thickness at the intact area of the straight pipe and the elbow pipe are 1.44% and 4.18%, respectively, and the differences of true and estimation thickness at the center of damage of the elbow pipe is 7.22%. However, the severity of the damage at the center of the damage region of the straight pipe was highly underestimated, which is 25.0% difference with the true thickness; the authors suspect it is due to the higher ratio of S_0 mode wavelength/true thickness in that case, which is $1.92 \text{ (cm)}/3.60 \text{ (cm)} = 53.3\%$, whereas for the case of the elbow pipe at the center of damage, this ratio is $2.27 \text{ (cm)}/8.03 \text{ (cm)} = 28.3\%$.

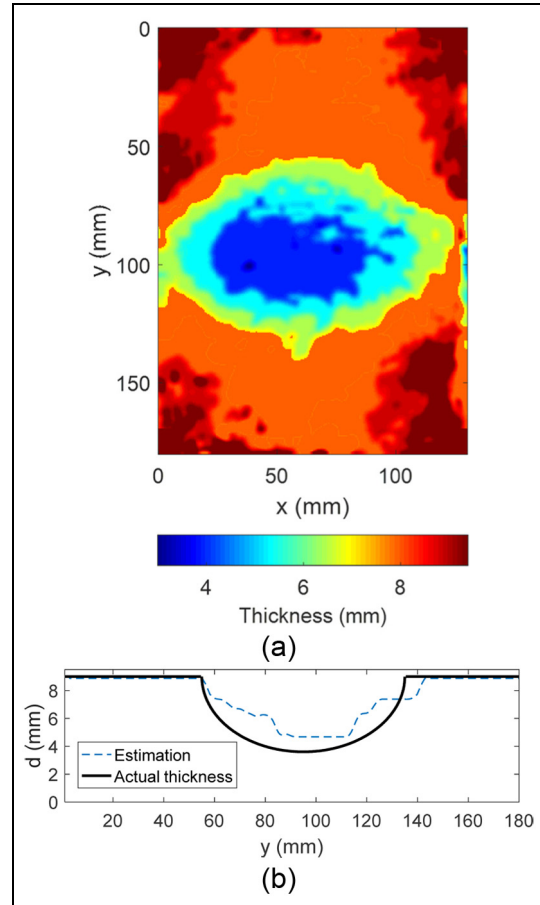


Figure 14. (a) Thickness reconstruction map of the straight pipe and (b) a slide across the thickness reconstruction map at $x = 65$ mm.

Discussion and conclusion

The FAC damage type is a special challenging case for damage visualization using the full-field scanning ultrasonic systems. The area-span type with smooth boundary of the FAC cannot trap a significant amount of wave energy inside the damage area, which renders many algorithms^{17,18,24} based on this phenomenon ineffectively. Even though the laser ultrasonic wavefield still can be used to detect the existent of the FAC damage, thanks to the reflected waves from the damage toward the sensor, it is unable to estimate the shape,

Table 1. Comparison of true and estimation of thickness of the straight and elbow pipes.

	Straight pipe			Elbow pipe		
	True value (mm)	Estimation (mm)	Difference (%)	True value (mm)	Estimation (mm)	Difference (%)
Intact region	9.0	8.87	1.44	11.0	11.46	4.18
Center of damage	3.60	4.50	25.0	8.03	8.61	7.22

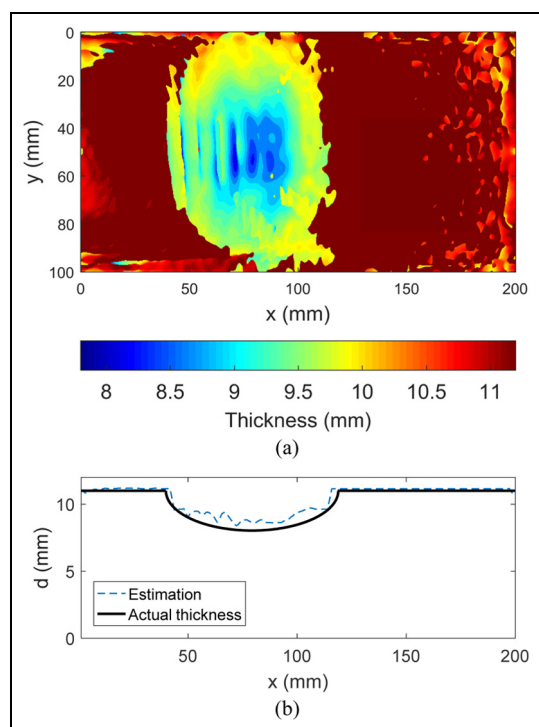


Figure 15. (a) Thickness reconstruction map of the elbow pipe and (b) a slide across the thickness reconstruction map at $y = 50$ mm.

size, or the severity of the damage. In this article, the FAC damages of steel pipes were effectively visualized using the UWI algorithm that maps the dominant local spectroscopic wavenumber of the wavefield at a particular mode and a particular frequency. Among many algorithms developed for processing full-wavefield data, the UWI algorithm has a unique approach: it provides the physical measurement quantity, rather than an arbitrary scaled intensity. One contribution of this article is a novel theoretical study to find the optimum parameters for UWI algorithms based on a sensitivity analysis of the spectroscopic wavenumber with respect to the change in thickness. This is a response to the open question posed in the previous wavenumber-processing contributions, calling for such a method. Finally, an inverse method was proposed to estimate the thickness of the structure based on the UWI map. From the analysis in this article, as a rule of thumb, when the specimen is thin, the wavenumber of the A_0 mode is more sensitive to thickness changes; however, when the specimen is thick, the S_0 mode is often more sensitive to thickness changes. In the future work, further investigation to reduce the artifact near the edges of the inspected area due to the curvature of the pipes is needed, such as by modifying the scanning scheme with the LMS. Also, real-world damage

characterization requires investigation with actual corrosion pipes instead of artificial corrosion, as in this article.

Declaration of conflicting interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This research was supported by Development of the Composite Lattice Structure (15-CM-MA-12) and Leading Foreign Research Institute Recruitment Program (2011-0030065), through the National Research Foundation of Korea, funded by the Ministry of Science, ICT, and Future Planning.

References

1. Zhang GA, Zeng L, Huang HL, et al. A study of flow accelerated corrosion at elbow of carbon steel pipeline by array electrode and computational fluid dynamics simulation. *Corros Sci* 2013; 77: 334–341.
2. Takahashi K, Tsunoi S, Hara T, et al. Experimental study of low-cycle fatigue of pipe elbows with local wall thinning and life estimation using finite element analysis. *Int J Pres Ves Pip* 2010; 87: 211–219.
3. Chen X, McLaury BS and Shiraz SA. Numerical and experimental investigation of the relative erosion severity between plugged tees and elbows in dilute gas/solid two-phase flow. *Wear* 2006; 261: 715–729.
4. Yurmanov V and Rakhmanov A. Flow accelerated corrosion of pipelines and equipment at Russian NPPs: problems and solutions. In: *IAEA workshop*, 2009, https://www.iaea.org/NuclearPower/Downloads/PLIM/2009-Apr-WS-Moscow/15_Yurmanov_ENES.pdf
5. Kastchiev G, Lyman E, Kromp W, et al. Residual risk: an account of events in nuclear power plants since the Chernobyl accident in 1986. Report commissioned by European Parliament, 2007, <http://large.stanford.edu/courses/2013/ph240/bechstein1/docs/kastchiev.pdf>
6. Findlan S, Munson D and Horowitz J. *Recommendations for an effective flow accelerated corrosion program*. EPRI Report NSAC-202L-R2, 2007. California, USA: Electric Power Research Institute.
7. Lee JR, Takatsubo J and Toyama N. Disbond monitoring at wing stringer tip based on built-in ultrasonic transducers and a pulsed laser. *Smart Mater Struct* 2007; 16(4): 1025–1035.
8. Staszewski WJ, Lee BC, Mallet L, et al. Structural health monitoring using scanning laser vibrometry: I. Lamb wave sensing. *Smart Mater Struct* 2004; 13(2): 251–260.
9. Ostachowicz W, Radziński M and Kudela P. Comparative studies of full wavefield signal processing for crack detection. *Strain* 2014; 50: 275–291.

10. Pei J, Yousuf MI, Degertekin FL, et al. Lamb wave tomography and its application in pipe erosion/corrosion monitoring. *Res Nondestruct Eval* 1996; 8(4): 189–197.
11. Cawley P and Alleyne D. The use of Lamb waves for the long range inspection of large structures. *Ultrasonics* 1996; 34(2): 287–290.
12. Bingham JP and Calvo DC. Propagation of low-frequency guided waves in three-dimensional submerged heterogeneous pipes. *J Acoust Soc Am* 2009; 126: 2188.
13. Chia CC, Lee JR and Shin HJ. Hot target inspection using a welded fibre acoustic wave piezoelectric sensor and laser-ultrasonic mirror scanner. *Meas Sci Technol* 2009; 20(12): 127003.
14. Lee JR, Jeong HM, Chia CC, et al. Application of ultrasonic wave propagation imaging method to automatic damage visualization of nuclear power plant pipeline. *Nucl Eng Des* 2010; 240(10): 3513–3520.
15. Scruby CB and Drain LE. *Laser ultrasonics techniques and applications*. New York: CRC Press, 1990.
16. Lee JR, Chia CC, Park CY, et al. Laser ultrasonic anomalous wave propagation imaging method with adjacent wave subtraction: algorithm. *Opt Laser Technol* 2012; 44(5): 1507–1515.
17. Sohn H, Dutta D, Yang JY, et al. Delamination detection in composites through guided wave field image processing. *Compos Sci Technol* 2011; 71(9): 1250–1256.
18. Ruzzene M, Jeong SM, Michaels TE, et al. Simulation and measurement of ultrasonic waves in elastic plates using laser vibrometry. *Rev Prog Q* 2005; 24: 172–179.
19. Chia CC, Lee JR and Park CY. Radome health management based on synthesized impact detection, laser ultrasonic spectral imaging, and wavelet-transformed ultrasonic propagation imaging methods. *Compos Part B: Eng* 2012; 43: 2898–2906.
20. Rogge MD and Johnston PH. Wavenumber imaging for damage detection and measurement. *AIP Conf Proc* 2011; 1430: 761–768.
21. Ruzzene M. Frequency–wavenumber domain filtering for improved damage visualization. *Smart Mater Struct* 2007; 16(6): 2116–2129.
22. Michaels TE, Michaels JE and Ruzzene M. Frequency–wavenumber domain analysis of guided wavefields. *Ultrasonics* 2011; 51(4): 452–466.
23. Yu L and Tian Z. Lamb wave structural health monitoring using a hybrid PZT-laser vibrometer approach. *Struct Health Monit* 2013; 12(5–6): 469–483.
24. Kudela P, Radziński M and Ostachowicz W. Identification of cracks in thin-walled structures by means of wavenumber filtering. *Mech Syst Signal Pr* 2015; 50–51: 456–466.
25. Rogge MD and Leckey CA. Characterization of impact damage in composite laminates using guided wavefield imaging and local wavenumber domain analysis. *Ultrasonics* 2013; 53(7): 1217–1226.
26. Mesnil O, Yan H, Ruzzene M, et al. Fast wavenumber measurement for accurate and automatic location and quantification of defect in composite. *Struct Health Monit* 2016; 15(2): 223–234.
27. Flynn EB, Chong SY, Jarmer GJ, et al. Structural imaging through local wavenumber estimation of guided waves. *NDT&E Int* 2013; 59: 1–10.
28. Truong TC, Kang DH, Lee JR, et al. Comparative study of laser Doppler vibrometer and capacitive air-coupled transducer for Ultrasonic Propagation Imager and the new development of an efficient ultrasonic wavenumber imaging algorithm. *Strain* 2015; 51(4): 332–342.
29. Koskelo EC and Flynn EB. Scanning laser ultrasound and wavenumber spectroscopy for in process inspection of additively manufactured parts. *Proc SPIE* 2016; 9804: 980418.
30. Truong TC and Lee JR. A versatile inspection system for pipe structure using ultrasonic waves propagation imager. *J Phys Conf Ser* 2015; 628(1): 012015.
31. Rose JL. *Ultrasonic waves in solid media*. Cambridge: Cambridge University Press, 1999.